Disclaimer/credit. These questions have largely been ~~stolen~~ borrowed from natural sciences maths and differential equations courses.

1) If $xyz + x^3 + y^4 + z^5 = 0$ (an implicit equation for any of the variables x, y, z in terms of the other two), find

$$\left(\frac{\partial x}{\partial y}\right)_z, \qquad \left(\frac{\partial y}{\partial z}\right)_x, \qquad \left(\frac{\partial z}{\partial x}\right)_y,$$

and show that their product is -1.

2) Van der Waals's equation

$$\left(p + \frac{a}{V^2}\right)(V - b) = RT$$

is an early (and in many ways a remarkably successful) attempt to represent the relation between the pressure p, volume V and temperature T of a real gas. (R, a and b are constants for a given mass of the gas.) Find expressions for $(\partial p/\partial V)_T$, $(\partial V/\partial T)_p$ and $(\partial T/\partial p)_V$, and verify that their product is -1.

3) $f(x, y)$ is a scalar function of position on the (x, y) plane. Position may also be specified by Cartesian coordinates u, v which are referred to axes rotated by angle θ from the x and y axes. Show that

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = \frac{\partial^2 f}{\partial u^2} + \frac{\partial^2 f}{\partial v^2},$$

i.e. the two-dimensional ∇^2 operator is invariant under a rotation of axes.

4) The enthalpy of a gas is defined by $H = U + pV$, where U satisfies $\mathrm{d}U = T\,\mathrm{d}S - p\,\mathrm{d}V$. Determine a relationship between the differentials of H, S and p. Hence show that

$$\left(\frac{\partial V}{\partial S}\right)_p = \left(\frac{\partial T}{\partial p}\right)_S.$$

By regarding U as a function of p and V and considering two expressions for $\partial^2 U/\partial p\,\partial V$, show that

$$\left(\frac{\partial S}{\partial V}\right)_p \left(\frac{\partial T}{\partial p}\right)_V - \left(\frac{\partial S}{\partial p}\right)_V \left(\frac{\partial T}{\partial V}\right)_p = 1.$$

5) Given that

$$\mathrm{d}U = T\,\mathrm{d}S - p\,\mathrm{d}V,$$

find a function G such that

$$\mathrm{d}G = V\,\mathrm{d}p - S\,\mathrm{d}T.$$

Hence show that

$$\left(\frac{\partial S}{\partial p}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_p.$$

6) The pressure p can be considered as a function of the variables V and T or as a function of the variables V and S.

(i) Find an expression for

$$\left(\frac{\partial p}{\partial V}\right)_T - \left(\frac{\partial p}{\partial V}\right)_S \quad \text{in terms of} \quad \left(\frac{\partial S}{\partial V}\right)_T \quad \text{and} \quad \left(\frac{\partial S}{\partial p}\right)_V.$$

(ii) Hence, using $T\,\mathrm{d}S = \mathrm{d}U + p\,\mathrm{d}V$, show that

$$\left(\frac{\partial \ln p}{\partial \ln V}\right)_T - \left(\frac{\partial \ln p}{\partial \ln V}\right)_S = \left(\frac{\partial (pV)}{\partial T}\right)_V \left[\frac{p^{-1}(\partial U/\partial V)_T + 1}{(\partial U/\partial T)_V}\right]. \qquad (\star)$$

$\left[\textit{Hint: } \left(\frac{\partial \ln p}{\partial \ln V}\right)_T = \frac{V}{p}\left(\frac{\partial p}{\partial V}\right)_T.\right]$

(iii) For one mole of an ideal gas, you may assume that $U = C_V T$, $pV = RT$, and pV^γ depends only on S, where C_V, R and γ are constants.
Use equation $(\star)$ to find an expression for γ in terms of C_V and R.

(iv) What is the value of γ for a monatomic gas for which $C_V = \frac{3}{2}R$?

7) For $f(x, y) = \exp(-xy)$, find $(\partial f/\partial x)|_y$ and $(\partial f/\partial y)|_x$. Check that

$$\frac{\partial^2 f}{\partial x\,\partial y} = \frac{\partial^2 f}{\partial y\,\partial x}.$$

Now express x and y in polar coordinates, $x = r\cos\theta$ and $y = r\sin\theta$. Find $(\partial f/\partial r)|_\theta$ and $(\partial f/\partial \theta)|_r$ using:

(a) the chain rule; and

(b) by first expressing f in terms of r and θ.

Check that the two methods give the same results.

8) Consider $F(s) \equiv \displaystyle\int_0^{1+s} f(x; s)\,\mathrm{d}x$.

(i) Give an expression for $\mathrm{d}F/\mathrm{d}s$.

(ii) If $f(x; s) = x(s - x)$, determine $\mathrm{d}F/\mathrm{d}s$ using your expression in (i).

(iii) For this $f(x; s)$, calculate the integral $F(s)$ directly and then differentiate with respect to s to confirm your answer to (ii).

(iv) For $f(x; s) = x(s - x)$, state the values of s for which $\mathrm{d}F/\mathrm{d}s = 0$.

9) Consider the change of variables

$$x = e^{-s}\sin t, \qquad y = e^{-s}\cos t,$$

such that $u(x, y) = v(s, t)$.

(i) Use the chain rule to express $\partial v/\partial s$ and $\partial v/\partial t$ in terms of x, y, $\partial u/\partial x$ and $\partial u/\partial y$.

(ii) Find, similarly, an expression for $\partial^2 v/\partial t^2$.

(iii) Hence transform the equation

$$y^2\frac{\partial^2 u}{\partial x^2} - 2xy\frac{\partial^2 u}{\partial x\,\partial y} + x^2\frac{\partial^2 u}{\partial y^2} = 0$$

into a partial differential equation for $v(s, t)$.

10) Note – you don't actually need to do the solution for this equation in this course, it has been left in as it could be an instructive application.

Solve the partial differential equation

$$\frac{\partial^2 u}{\partial x^2} + 2\frac{\partial^2 u}{\partial x\,\partial y} + \frac{\partial^2 u}{\partial y^2} = 0 \qquad (*)$$

for $u(x, y)$ by making a change of variables as follows. Define new variables

$$\xi = x - y, \qquad \eta = x,$$

and evaluate the partial derivatives of x and y with respect to ξ and η. Writing $v(\xi, \eta) = u(x, y)$, use these derivatives and the chain rule to show that

$$\frac{\partial v}{\partial \eta} = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y},$$

and that the equation

$$\frac{\partial^2 v}{\partial \eta^2} = 0$$

is equivalent to $(*)$.

Deduce that the most general solution of $(*)$ is

$$u(x, y) = f(x - y) + x\, g(x - y),$$

where f and g are arbitrary functions.

Solve $(*)$ completely given that $u(0, y) = 0$ for all y, whilst $u(x, 1) = x^2$ for all x.